

# MongoDB Coding Assessment

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25/07/25

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## Relations In MongoDB:

MongoDB is a NoSQL, document-based database—it stores data in JSON-like documents (BSON), not tables. While it is schema-less and doesn't use traditional relational models like SQL databases, you can still represent relationships between data in three main ways:

### 1. One-to-One (1:1):

```
db.users.insertOne({
  _id: 1,
  name: "Aarthi",
  email: "12345@example.com",
  profile: {
    age: 21,
    gender: "Female",
    bio: "Data enthusiast"
  }
})
{
  acknowledged: true,
  insertedId: 1
}
db.users.find()
{
  _id: 1,
  name: 'Aarthi',
  email: '12345@example.com',
  profile: {
    age: 21,
    gender: 'Female',
    bio: 'Data enthusiast'
  }
}
```

## 2. One-to-Many (1:N):

```
> db.users.insertOne({
  _id: 10,
  name: "Aarthi",
  email: "12345@example.com",
  profile: {
    age: 21,
    gender: "Female",
    bio: "Data enthusiast"
  },
  posts: [
    { post_id: 201, title: "MongoDB Basics", content: "Learned about NoSQL." },
    { post_id: 202, title: "Data Modeling", content: "Exploring relations in MongoDB" }
  ]
})
< {
  acknowledged: true,
  insertedId: 10
}
```

```
> db.users.find().pretty()
< {
  _id: 10,
  name: 'Aarthi',
  email: '12345@example.com',
  profile: {
    age: 21,
    gender: 'Female',
    bio: 'Data enthusiast'
  },
  posts: [
    {
      post_id: 201,
      title: 'MongoDB Basics',
      content: 'Learned about NoSQL.'
    },
    {
      post_id: 202,
      title: 'Data Modeling',
      content: 'Exploring relations in MongoDB'
    }
  ]
}
```

### 3. Many-to-Many (N:N):

```
> db.users.insertOne({
  _id: 11,
  name: "Aarthi",
  email: "12345@example.com",
  profile: {
    age: 21,
    gender: "Female",
    bio: "Data Enthusiast"
  },
  enrolled_courses: [301, 302]
})
< {
  acknowledged: true,
  insertedId: 11
}
> db.createCollection("courses")
< { ok: 1 }
> db.courses.insertMany([
  { _id: 301, name: "Data Modelling", enrolled_user_ids: [11, 12] },
  { _id: 302, name: "Data Warehousing", enrolled_user_ids: [11] }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': 301,
    '1': 302
  }
}
}
```

```
> db.users.find().pretty()
< {
  _id: 11,
  name: 'Aarthi',
  email: '12345@example.com',
  profile: {
    age: 21,
    gender: 'Female',
    bio: 'Data Enthusiast'
  },
  enrolled_courses: [
    301,
    302
  ]
}

> db.courses.find().pretty()
< {
  _id: 301,
  name: 'Data Modelling',
  enrolled_user_ids: [
    11,
    12
  ]
}
{
  _id: 302,
  name: 'Data Warehousing',
  enrolled_user_ids: [
    11
  ]
}
```